

A System One model inside Claude Code

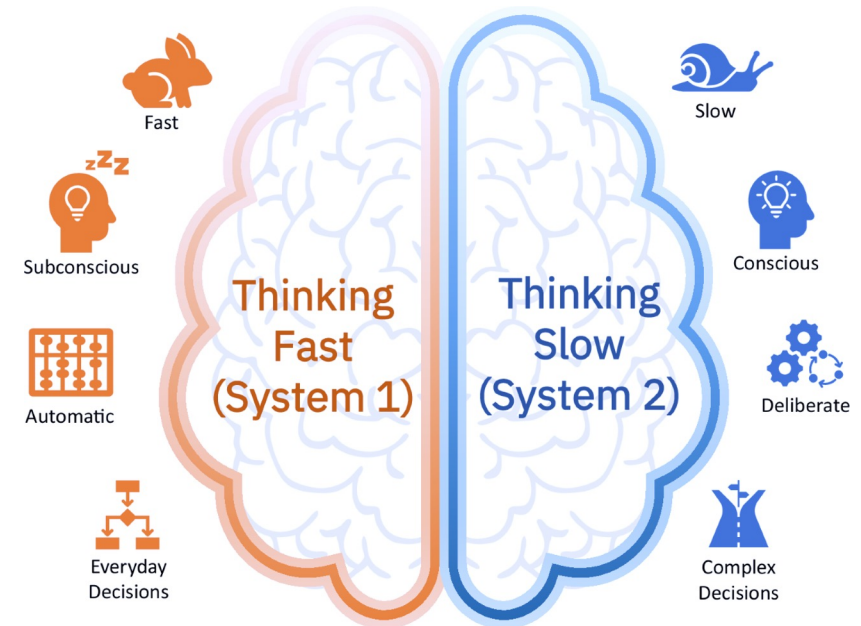
Early experience with Jev: three hooks, three benchmarks, one live demo

SYSTEM 1

answers at once, with a probability

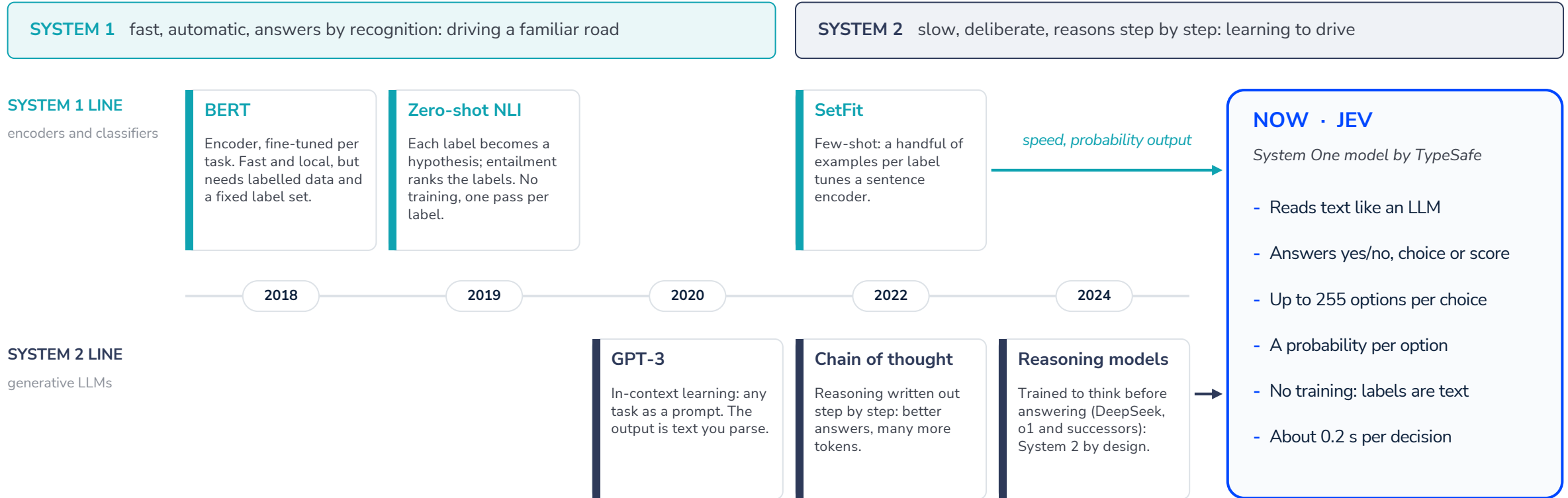
SYSTEM 2

thinks, then answers



From BERT to System One

Two lines of language models, fast classifiers and slow reasoners, and where Jev sits.



Jev joins both lines: LLM-level reading with a classifier's output you can put in an if statement.

What people are already building with Jev

Four public experiments, one pattern: select, don't generate. Jev picks, ordinary code builds.



Where System One fits: 2 industrial patterns

Many small decisions over a stream of text, with a label set that keeps changing.

Retrieval-augmented generation (RAG)

Which knowledge should this answer rest on?

Which source should answer this query?

CHOICE

Route the query to one index, or to several above a cut-off.

Does this chunk answer the question?

YES / NO

Filter retrieved chunks; 240 questions fit in one request.

Is the draft answer supported by its chunks?

SCORE

Release it, or escalate to an LLM or a human.

WHY SYSTEM ONE HERE

It runs on every chunk of every query, so price and latency decide the design.

Manufacturing log classification

What is this log line, and who needs to see it?

Which fault class is this line?

CHOICE

Up to 255 described classes; new classes need no retraining.

How urgent is it for the on-call engineer?

SCORE

Each severity level is described in words; page or batch.

Does it match a known issue in the runbook?

YES / NO

Attach the runbook entry; unknown patterns go to a human.

WHY SYSTEM ONE HERE

High volume, and labels that move with the plant: a probability sets the cut-off.

Same shape in both: many small decisions, labels that change, and a probability that sets the cut-off.

Real Manufacturing use case : Failure Modes hidden in maintenance logs

One notification: a few structured fields, and the real evidence in short French texts.

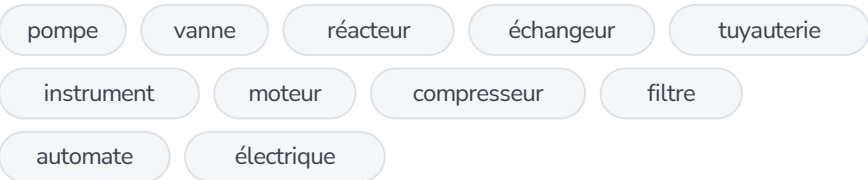
SAP PM notification (corrective)		ILLUSTRATIVE
Description	Fuite garniture pompe de transfert	
Technical post	P-101	
Equipment type	Pompe centrifuge	
Priority / ABC class	2 / A	
Technician's report	Fuite de produit à la garniture mécanique, goutte à goutte. <i>product leaking at the mechanical seal, drop by drop</i>	
Shift lead's check	Confirmé par le chef de poste, pompe de secours en service. <i>confirmed by the shift lead, standby pump running</i>	
Planning note	Prévoir garniture de rechange et consignation électrique. <i>plan a spare seal and electrical lock-out</i>	
+ 2 more text fields	Specification note, free comment	

WHAT WE NEED FROM IT

179

failure-mode codes to choose from, each with a French description and definition

ACROSS 11 EQUIPMENT CATEGORIES



OUTPUT

Up to two codes per notification, each with a confidence and a short reason, so a reliability engineer can trace every label back to the text.

The label is one choice among 179 codes, read from five short French texts: a System 1 shape.

How we would test it: A System 1 pilot on the same data

Five small questions per notification cover the labelling job and the bad-actor inputs.

	JEV QUESTION	FEEDS
CHOICE	Which failure mode does this notification describe? <i>179 described codes</i>	The failure-mode label for the backlog
CHOICE	Which failure-mode family does it belong to? <i>the failure-mode families</i>	A consistency check: the code must sit in its family
YES / NO	Do these two notifications describe the same failure on the same asset? <i>pairs per technical post</i>	Recurrence and time between failures
CHOICE	Which Ishikawa 6M category fits the likely cause? <i>6 options</i>	Input to the root-cause workshop
SCORE	How severe was the impact on production? <i>levels described in words</i>	Weights the Pareto beyond Priority and ABC

Closest evidence we have: in our memory write gate, Jev judged 'same fact?' right on 189 of 189 held-out pairs.

One request per notification answers all five; labelling and bad-actor inputs come out together.

Hands-On Tests: Jev vs Haiku 4.5 vs Local models

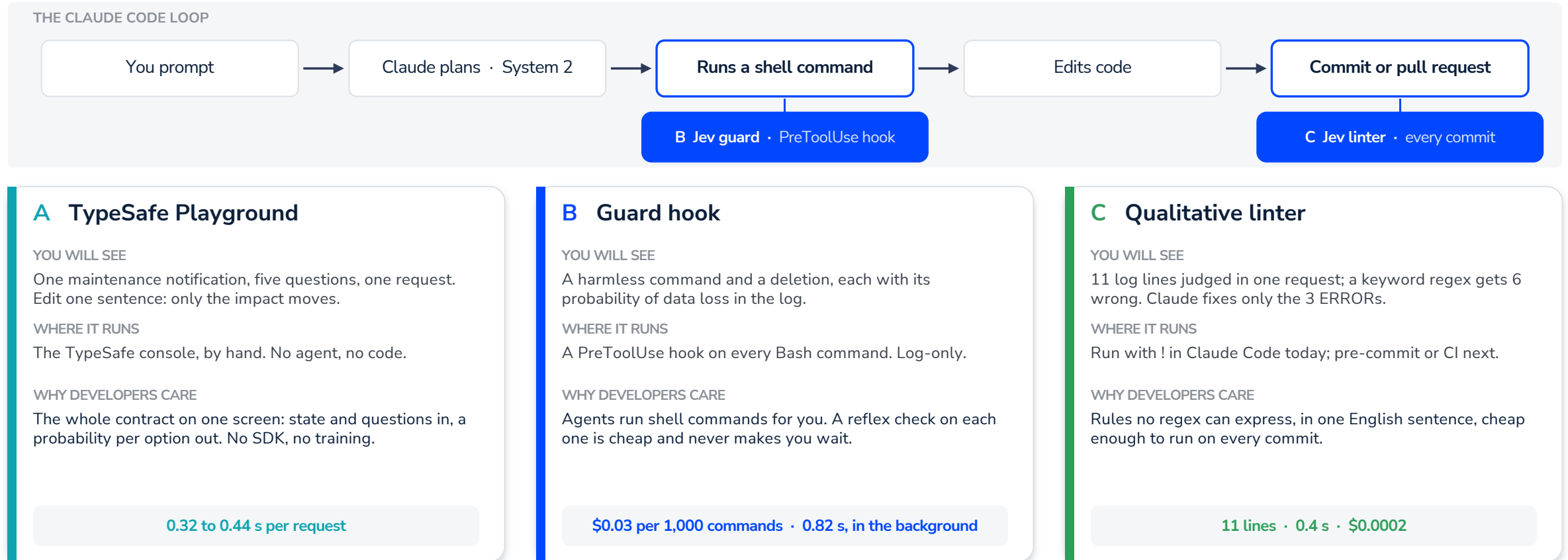
I ran my own tests comparing performance of Jev with Local Alternatives as well as Haiku (thinking off)

PER HOOK	Jev jev-1.13.0, TypeSafe API	Haiku 4.5 thinking off, Anthropic API	reflex 4B Qwen3.5-4B, this Mac, 9.3 GB	reflex 2B Qwen3.5-2B, this Mac, 4.55 GB	VERDICT
COMMAND GUARD · hard held-out set, 99 commands					Jev and Haiku tie on accuracy; Jev costs 1/36. Local models flag 25 and 43 of 53 safe commands: not usable.
Both right, two yes/no questions	82	90	63	53	
Both right, one 0-4 loss score	88	-	53	45	
Destroys missed / false alarms	3 / 7	2 / 5	6 / 25	3 / 43	
Median time per command	0.32 s	0.91 s	3.41 s	1.35 s	
Cost per 1,000 commands	\$0.03	\$1.09	\$0	\$0	
PROMPT ROUTER · 86 skills, 63 tools					Jev wins on clear prompts and never invents a name. Both are weak on real, vague prompts. No local run: 26-option limit.
Constructed, top 1 / top 5 of 133	122 / 130	103 / 118	not run	not run	
Real prompts, top 1 / top 5 of 26	12 / 15	14 / 16	-	-	
Time, cost per 1,000 prompts	1.09 s, \$0.41	2.35 s, \$2.50	-	-	
MEMORY WRITE GATE · 189 held-out pairs					Jev near perfect in one batched call. reflex 4B is close enough for this job alone, at 2.9 s per pair.
Right per question, of 189	189/186/174	waived	186/182/153	152/139/131	
Throughput	60 pairs, 1.7 s	-	1 pair, 2.9 s	1 pair	

Latest rankings on JeVBench can be found here : <https://benchmarkheaven.com/jev-models> & <https://jevbench.dev/leaderboard>

Over to the terminal

A coding agent makes thousands of small decisions. Today two of them get a reflex.



The agent is the thinker, Jev is its reflex where a check is cheap to run and costly to miss.

Thank you

Before you call an LLM, ask : Does this decision need a thinker, or a reflex ?

THE DEMO, READY TO REPLICATE

github.com/uguremrah/jev-experiments

The Playground case, the qualitative linter, measured results. MIT licence.

